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Compact and broadband polarization beam splitter based on a bridged hybrid plasmonic waveguide coupler on a thin-film lithium niobate

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We propose a compact and broadband polarization beam splitter (PBS) based on a thin-film lithium niobate. The hybrid plasmonic waveguide (HPW) is used as a bridge in a three-waveguide coupler to achieve large birefringence. A degree of phase-matching is introduced for optimization. The proposed PBS exhibits a 125-nm bandwidth (1495–1620 nm) of polarization extinction ratio (PER) over 20 dB or an 85-nm bandwidth (1515–1600 nm) of PER over 25 dB, with a device length of only 81 μm . An ultra-low insertion loss of 0.04 dB is achieved around the wavelength of 1550 nm. In addition, the proposed device exhibits high robustness of the fabrication tolerance. © 2024 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

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Lithium niobate (LN) is a well-known material in integrated photonics for its excellent electro-optic, acousto-optic, and non-linear optical properties. Over the past years, LN has gained renewed interest, thanks to the commercial availability of thin-film LN-on-insulator (LNOI) wafers that offer sub-wavelength light confinement [1], enabling the development of a wide range of increasingly compact integrated devices, including electro-optical modulators, high-quality optical cavities, and broadband frequency comb sources [2–6]. Many devices based on the LNOI platform are polarization-dependent due to geometric birefringence and inherent anisotropy of the LN waveguides. To address this problem, the polarization diversity scheme has been proposed [7,8].

The polarization beam splitter (PBS) is one of the key components in the polarization diversity circuit for separating or combining the transverse-electric (TE) and transverse-magnetic (TM) polarization beams. Moreover, PBSs are also widely used in polarization-division multiplexing and polarization entanglement quantum systems [9,10]. Previously, various schemes have been proposed to implement PBSs on integrated platforms, including multimode-interference (MMI) couplers, Mach-Zehnder interferometers (MZIs), sub-wavelength grating (SWG) metamaterials, and asymmetric directional couplers (ADCs) [11–17]. Most of the devices are based on the

silicon-on-insulator (SOI) platform, where giant birefringence can be achieved by high index contrast and small cross section (e.g., 220 nm). However, LNOI waveguides have moderate index contrast and usually require a relatively thick layer (e.g., 300–900 nm), resulting in low structural birefringence. Therefore, the general design of MMI- or DC-based PBSs leveraging distinct self-image beat length or evanescent-coupling strength between TE and TM modes are not suitable for LNOI platform. The MZI and SWG structures are promising candidates to implement broadband PBSs on the LNOI platform. Xu *et al.* proposed an MZI-based PBS with excellent results [13]. However, the device size is in sub-millimeter range due to the cascaded structure of the phase shifter. Ref. [15] proposed a SWG-based PBS via engineering the artificial anisotropy of SWGs to improve polarization selectivity, but it is a challenge to fabricate such devices on LNOI platform due to small feature sizes.

The ADC-based PBS is the most common type for its simple structure and small footprint. The idea is to assemble waveguides of distinct birefringence and to meet the phase-matching condition for the polarization by design. Particularly, the three-waveguide coupler is an alternative to achieve PBSs with high extinction ratio and large bandwidth [17–22]. Hybrid plasmonic waveguides (HPWs) [18,19], Si- or SiN-assisted waveguides [20,21], and nano-slot waveguides [22] are employed as the bridge in the three-waveguide coupler. Basically, the TM-mode coupling is preferred due to stronger evanescent-coupling, especially on the SOI platform. However, the coupling strength of the TE and TM modes can be comparable on the LNOI platform as a result of relatively low birefringence. Additionally, in the heterogeneous loading schemes [18–21], polarized beams of different transverse directions interact with the vertically loading material distinctly, and thus their wavelength dependence can be largely different. The TM-mode coupled PBSs based on a directional coupler on the LNOI platform exhibit moderate bandwidth and small fabrication tolerance [19,21].

In this Letter, we propose a TE-mode coupled PBS based on a bridged HPW coupler on an X-cut LNOI platform with broad working bandwidth, low insertion loss, and large fabrication tolerance. We introduce the degree of phase-matching to estimate the wavelength dependence and fabrication tolerance and modify the waveguide dimensions to achieve the optimum overall performance with a 125-nm bandwidth of PER >20 dB or an

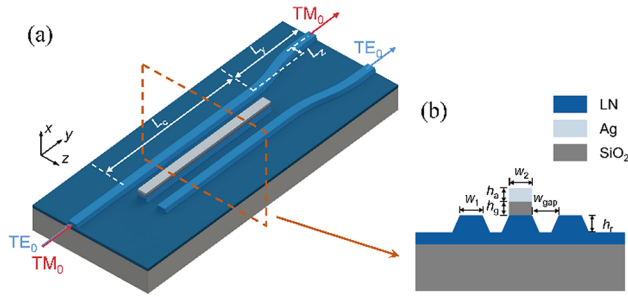


Fig. 1. Schematic structure of the PBS. (a) Three-dimensional view and (b) cross section view.

85-nm bandwidth of PER >25 dB, an IL of 0.04 dB at the central wavelength for both modes. Simulation shows fabrication tolerance is expected to be greater than ± 40 nm, making it easy for manufacturing process in the future.

Figure 1(a) shows a schematic diagram of the proposed PBS based on an X-cut LNOI wafer, which has a 500-nm-thick LN layer and a 3- μm -thick buried oxide (BOX) layer. The device is designed to separate the two orthogonal polarization components into two output ports, i.e., through port for fundamental TM mode and cross port for fundamental TE mode. As depicted in Fig. 1(b), the coupling section of the PBS is composed of two identical single-mode waveguides (SMWs) and an HPW. The latter consists of an LN ridge waveguide, a silica buffer, and an Ag cap. In this Letter, the simulated sidewall angle θ is fixed at 72° according to the current fabrication technology, and the LN ridge height, denoted as h_r , is set to be 300 nm. The propagation direction of the waveguides is along the crystal y axis of LN, which is compatible with most electro-optic modulators on the LNOI platform [2,3].

The Finite-Difference Eigenmode (FDE) solver (Ansys Lumerical MODE Solutions) is utilized to analyze the electric field profiles of modes in the SMW and HPW. The waveguide width is set to be 500 nm to meet the single-mode condition, and the thickness of the Ag cap is chosen as $h_a = 100$ nm in consideration of easy fabrication. In Fig. 2, one can see both the TE mode and TM mode are well confined in the SMW and have a similar refractive index of 1.721 and 1.660, respectively. However, in the case of HPW (e.g., $h_g = 160$ nm for the silica buffer), the TM mode is highly confined in the buffer layer, while the TE mode remains almost unchanged. The refractive index of the TE mode and the TM mode is 1.716 and 1.745, respectively. Based on the polarization-dependent effect of the metal cap, it is indicated

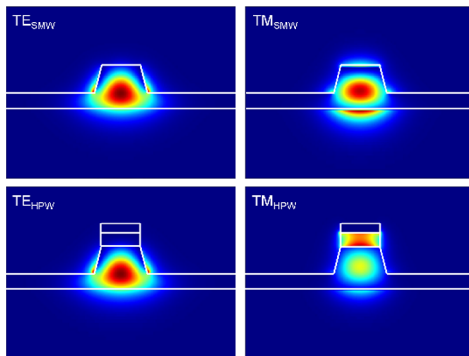


Fig. 2. Electric field profiles of TE and TM modes in the SMW and the HPW.

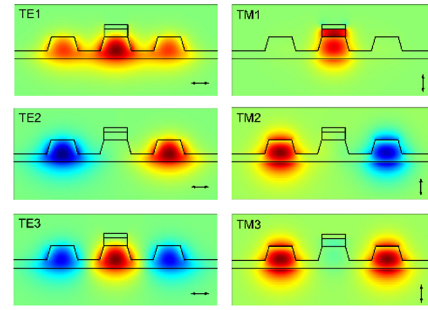


Fig. 3. Supermodes supported in the coupling section of the proposed PBS.

that the dispersion characteristics of the TE mode is less subject to the Ag cap, and thus the TE-coupled scheme is preferred.

Figure 3 shows the supermodes supported in the three-waveguide array. For the TE-mode input, three supermodes are excited, i.e., two even modes with propagation constants of β_{TE1} and β_{TE3} , and an odd mode with the propagation constant of β_{TE2} . To achieve maximum cross coupling, the phase-matching condition among the three modes can be expressed as follows [23]:

$$\Delta\beta = \beta_{TE1} + \beta_{TE3} - 2\beta_{TE2} = 0. \quad (1)$$

For the TM-mode input, the three-waveguide array acts like a two-waveguide DC since only the odd mode and even mode can be excited [18], with propagation constants of β_{TM2} and β_{TM3} . Therefore, cross coupling of the TM mode will not be negligible if the waveguide separation is not enough. Here, the gap width between the SMW and the HPW is chosen as $w_{gap} = 800$ nm to obtain a trade-off between device size and extinction ratio. The length of single complete coupling for the two modes can be derived as follows [18]:

$$L_{c,TE} = \frac{2\pi}{\beta_{TE1} - \beta_{TE3}}, \quad (2a)$$

$$L_{c,TM} = \frac{\pi}{\beta_{TM2} - \beta_{TM3}}. \quad (2b)$$

The thickness of the silica buffer, h_g , is further optimized in consideration of the high extinction ratio, broad working bandwidth, and large fabrication tolerance. Figure 4(a) depicts the contour plots of the degree of phase-matching, i.e., $\Delta\beta$, with the thickness varying from $h_g = 40$ nm to $h_g = 200$ nm and the bridge width ranging from $w_2 = 500$ nm to $w_2 = 570$ nm at the wavelength of $1.55 \mu\text{m}$. The black line represents the result of $\Delta\beta = 0$. Therefore, w_2 can be decided according to the value of h_g .

For the proposed PBS, a larger $L_{c,TM}/L_{c,TE}$ is preferred so that the TM mode achieves weaker coupling, leading to a higher PER. One can see that $L_{c,TM}/L_{c,TE}$ decreases with the increase of h_g in Fig. 4(b). Detailed characteristics of the TE-mode coupling are analyzed then. Figures 4(c) and 4(d) show the degrees of phase-matching as functions of wavelength and width deviation. One can see in Fig. 4(c) that the larger the h_g is, the more perfect the phase-matching condition is when the wavelength is off-center. We assume a deviation of Δw for both w_1 and w_2 , which means the opposite deviation for the gap width, i.e., $\Delta w_{gap} = -\Delta w$. Figure 4(d) shows minimum phase-mismatching can be obtained around $h_g = 160$ nm, indicating a large fabrication tolerance. Based on the above findings, we select the silica buffer thickness and the HPW bridge width of 160 and

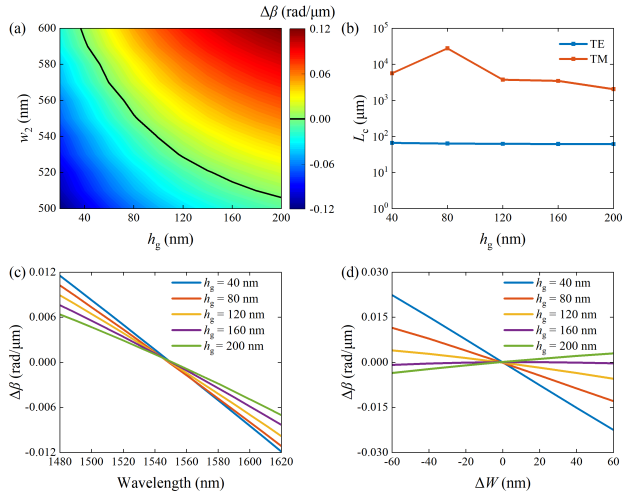


Fig. 4. (a) Contour plots of $\Delta\beta$ with variations in h_g and W_2 , (b) $L_{c, TE}$ and $L_{c, TM}$ as functions of h_g , (c) $\Delta\beta$ with variations in h_g and wavelength, and (d) $\Delta\beta$ with variations in h_g and ΔW .

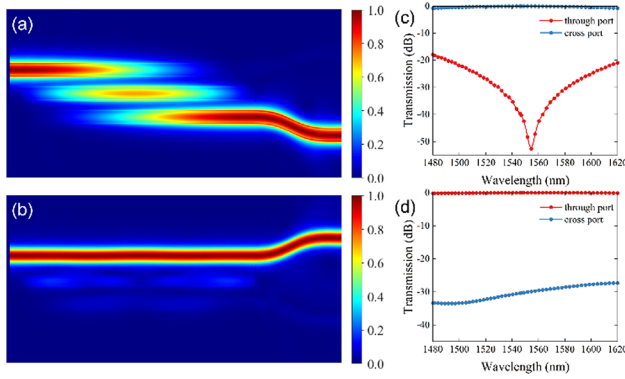


Fig. 5. Beam propagation profiles of (a) TE and (b) TM input at the wavelength of 1550 nm; the transmission spectra of (c) TE and (d) TM input.

512 nm, respectively. The coupling length of the TE mode is 61 μm . After the coupling section, s-bend waveguides are introduced to avoid further coupling. The footprint of the s-bend waveguide in our design is $1 \times 20 \mu\text{m}^2$. The optical transmission characteristics of the PBS are examined in detail using the 3D finite-difference time-domain (FDTD) software (Ansys Lumerical FDTD Solutions). Figures 5(a) and 5(b) show the electric field distribution for the TE and TM input at the wavelength

of 1.55 μm , respectively. One can see the coupling of the TE mode occurs in the three-waveguide array and completes at the entrance of the s-bend, while the TM mode propagates along the through port with negligible crossing due to its long coupling length.

Figures 5(c) and 5(d) show the transmission of the proposed PBS in the wavelength range of 1.48–1.62 μm with the TE and TM input, respectively. PER of 52.6 and 29.9 dB and insertion loss (IL) of 0.03 and 0.04 dB can be obtained for the TE and TM modes, respectively. For the TE mode, the bandwidth of PER > 20 dB is 125 nm (1495–1620 nm) with IL < 0.80 dB, or the bandwidth of PER > 25 dB is 85 nm (1515–1600 nm) with IL < 0.42 dB. For the TM mode, it can be observed in Fig. 5(d) that the PER is larger than 27.2 dB and IL is smaller than 0.17 dB over the whole wavelength range. Here, the PERs are defined as follows:

$$\text{PER}_{\text{TE}} = 10 \log_{10}(T_{\text{cross, TE}}/T_{\text{through, TE}}), \quad (3a)$$

$$\text{PER}_{\text{TM}} = 10 \log_{10}(T_{\text{through, TM}}/T_{\text{cross, TM}}). \quad (3b)$$

and ILs are given by

$$\text{IL}_{\text{TE}} = -10 \log_{10}(T_{\text{cross, TE}}), \quad (4a)$$

$$\text{IL}_{\text{TM}} = -10 \log_{10}(T_{\text{through, TM}}). \quad (4b)$$

Table 1 lists several recent simulations on PBSs based on the LNOI platform. Our PBS has the largest bandwidth of PER > 25 dB for the TE and TM modes combined, and the lowest IL at the central wavelength among all the listed PBSs. In addition, the device size is below 100 μm , which meets the mainstream level.

The fabrication tolerance of the proposed PBS is further investigated. A deviation of ± 40 nm in waveguide width is studied. It can be found in Figs. 6(a) and 6(b) that for the TE mode, the PER almost stay stable though the blueshifting appears with the deviation of $\Delta w = +40$ nm. But it is still over 30 dB around the central wavelength. One can see in Figs. 6(c) and 6(d) that a deviation of ± 20 nm in mask alignment causes the degradation of the PER for the TE mode, but it is still greater than 30 dB at the central wavelength, and there is no significant change for the bandwidth of PER > 20 dB.

In Figs. 6(e) and 6(f), a ± 10 nm deviation of the etching depth is also studied. For the TE mode, the redshifting and blueshifting occur but it is still more than 30 dB at the wavelength of 1.55 μm . In addition, a ± 10 nm deviation of buffer thickness is considered, as shown in Figs. 6(g) and 6(h). The transmission of the TE mode is more sensitive to the minus deviation, for the reason that the TE mode has a stronger interaction with the metal with the decrease of buffer thickness.

Table 1. Comparison of PBSs Based on the LNOI Platform^a

Structures	Cut	BW ₂₅ dB (nm) TE/TM	PER (dB) TE/TM	IL _{entire} (dB) TE/TM	IL (dB) TE/TM	Length (μm)	Year
MZI [13]	X-cut	80/140	47.7/48	<2.0/<1.7	0.9/0.6	430	2020
SWG [15]	N.A.	10/40	26/31	<1.0	<1.0	160	2021
Cascaded mode converters [24]	Z-cut	140/22	35/37	1.10/1.24	<0.15	180	2022
HPW (two-waveguide) [25]	X-cut	90/47	40.9/26.1	<0.60	<0.27	47	2023
HPW (adiabatic passage) [26]	Z-cut	35/140	22/43	<1.0	N.A./0.3	6000	2022
SiN-assisted coupler [21]	Z-cut	140/30	32/55	N.A.	<0.07/<0.16	31	2023
HPW (three-waveguide) [19]	Z-cut	100/18	28.3/31.7	N.A.	0.06/0.31	33	2024
This work	X-cut	85/140	52.6/29.9	<0.80/<0.17	0.03/0.04	81	2024

^aN.A., not available. BW₂₅ dB and IL_{entire} are the bandwidth for PER > 25 dB and insertion loss within the wavelength range of 1480–1620 nm (S, C, and L bands), respectively. PER and IL are the polarization extinction ratio and insertion loss around the central wavelength, respectively.

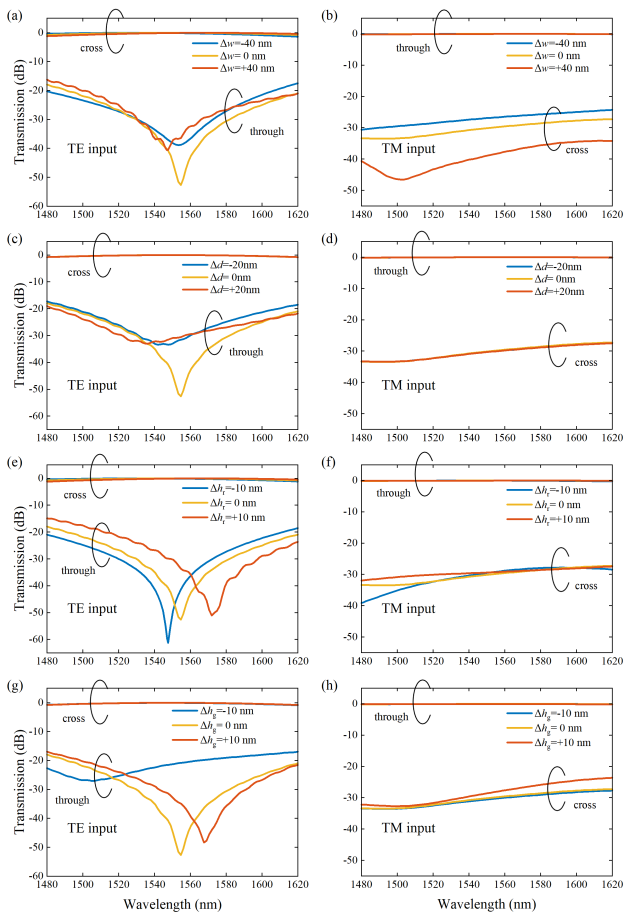


Fig. 6. (a)–(h) Tolerance analysis of deviations of waveguide width, mask alignment, etching depth, and buffer thickness.

In conclusion, a compact and broadband PBS based on a bridged HPW coupler on the LNOI platform has been proposed. With the help of the Ag-capped HPW, the three-waveguide coupler of our PBS allows for maximum coupling for the TE mode while suppressing the power exchange for the TM mode. The simulation shows that the PER of the PBS is 52.6 and 29.9 dB at the central wavelength for TE- and TM-mode inputs, respectively. The device exhibits a 125-nm bandwidth of PER >20 dB, covering C, L bands, and part of the S band. The bandwidth of PER >25 dB for the TE mode and the TM mode is 85 and 140 nm, respectively. An ultra-low IL of 0.04 dB is

achieved around the central wavelength for both modes. Additionally, our PBS exhibits high robustness, making it easy to be obtained by current fabrication technology. Our work provides, to our knowledge, a new strategy for polarization beam splitting, paving the ways for the polarization diversity applications in the integrated platforms of relatively low birefringence.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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